RESEARCH PROPOSAL

Human-Centered AI for Elderly Care & Remote Health Monitoring

A Privacy-Preserving, Multilingual System for Rural and Semi-Urban India

Abstract

India's population is ageing rapidly, alongside a rise in chronic illness and persistent gaps in healthcare access and resource distribution. Older adults currently make up roughly 7–8% of India's population; this share is projected to reach 20–21% by 2050 (UNFPA India & IIPS, 2023). This shift will make community-oriented healthcare for the elderly both more necessary and harder to deliver.

This proposal outlines the design and evaluation of a human-centered artificial intelligence system for elderly care and remote health monitoring. The system will use smartphone and wearable-device data to identify falls, abnormal vital signs, and prolonged inactivity. A caregiver dashboard will translate this data into understandable risk scores and prioritized alerts, while a simplified interface will support elderly users with limited digital literacy.

The study will draw lessons from Japan's technology-assisted elderly-care approaches, including monitoring, care-worker support, and assistive technologies, but will adapt them to India's economic, linguistic, infrastructural, and social conditions. A pilot evaluation will be conducted with hospitals, community health workers, NGOs, or elderly-care organizations in two or three Indian districts.

Literature Review

Population Ageing and the Indian Elderly-Care Context

Population ageing is becoming an increasingly important issue in India. As the elderly population grows, so does the need for better healthcare, financial support, and social care for older adults. Elderly people in India have traditionally relied on family-based support, but urbanisation, migration, smaller households, and changing lifestyles are eroding this support in many cases, contributing to loneliness and social isolation. Older adults commonly face multiple, overlapping health problems, including diabetes, hypertension, heart disease, mobility issues, and memory-related conditions. India's hospital-centric healthcare system, however, is built around treating individual diseases rather than providing the continuous, coordinated care that these overlapping, long-term conditions require (Vision IAS, 2026).

This context strengthens the case for home-based monitoring and community-level care coordination. However, a technology solution must account for differences in digital literacy, connectivity, device ownership, family support, and access to healthcare. The proposed research therefore treats affordability, usability, and caregiver workflow as core design requirements rather than secondary features.

Remote Monitoring and Telehealth for Older Adults

Research on telehealth for rural older adults indicates that it can improve access and support health promotion, although evidence on ease of use and continued usage is mixed. A systematic mixed-studies review found that telehealth was broadly useful across most included studies, though usability and cost-effectiveness varied (Kruse et al., 2021).

A ScienceDirect impact analysis of telemedicine in India projects that remote patient monitoring, including wearable trackers, smart glucose monitors, and connected BP cuffs, will become increasingly central to elderly and chronic-disease care, with the Ayushman Bharat Digital Mission (ABDM) expected to strengthen the underlying digital-health infrastructure over the next few years (ScienceDirect, 2025).

Fall Detection and Wearable/Sensor-Based AI

Falls remain one of the most researched and urgent problems in elderly-care technology, especially as India's elderly population is projected to reach around 12% of the national population, with 8–10% requiring serious ongoing care.

ICMR data further confirm that fall-related injuries among Indian seniors carry serious health consequences.

Most detection systems rely on wearable accelerometers and gyroscopes feeding data into machine-learning models. A 2025 study proposed LTC-Net, a dual-branch model combining dynamic feature extraction with temporal convolutional modeling to reduce false alarms.

A Chennai-based team achieved 97% accuracy using a CNN-LSTM model with attention layers on edge-computing hardware. Another IoT-based system reported 99% detection accuracy using LSTM models while also addressing energy efficiency and connectivity loss.

A CNN-LSTM model combines convolutional neural networks, used for spatial feature extraction, with long short-term memory networks, used for modeling temporal dependencies, allowing it to process data with both spatial structure and time-series dynamics, such as video, multivariate sensor streams, or audio.

Beyond detection, a Pune-based study compared sensor-based fall-risk assessment (Kinesis QTUG) against conventional clinical tools across 659 community-dwelling older adults over 12 months, providing a strong India-specific, real-world reference point.

Broader surveys frame this work under “Ambient Assisted Living Systems,” while acknowledging persistent challenges: false alarms, device placement, and low real-world user acceptance.

Explainable and Human-Centered AI in Healthcare

High-performing AI is not automatically suitable for healthcare. Explainability, transparency, uncertainty, and human oversight all influence whether users can appropriately rely on AI outputs. Loh et al. (2022) reviewed explainable AI applications in healthcare and identified the black-box nature of many AI models as a barrier to practical adoption.

A systematic review by Rosenbacke et al. (2024) further found that explanations can increase trust when they are clear, concise, and clinically relevant, but that complex or contradictory explanations can reduce trust.

Recent human-factors research also emphasizes calibrated trust: healthcare professionals and patients need neither blind trust nor unnecessary rejection of AI recommendations. Instead, systems should communicate their limitations and support human judgment (Choudhury & Shamszare, 2026).

This supports the proposed risk-score design, in which an alert will explain the principal evidence behind the risk category and require human verification before high-impact action is taken.

Japan and Technology-Assisted Elderly Care

Japan provides a valuable comparative setting because its ageing population has encouraged sustained development of, and policy support for, care technologies. Japan's Ministry of Health, Labour and Welfare, together with the Ministry of Economy, Trade and Industry, has promoted priority areas for care technology with goals including improved care quality, reduced staff burden, and greater independence for older people (MHLW, 2024).

The Japanese experience is particularly relevant to this study because it demonstrates that elderly-care technology should be connected to care workflows rather than treated only as a technical product. Direct transfer to India would nonetheless be inappropriate, given substantial differences in household structure, workforce availability, language diversity, connectivity, and affordability.

The proposed Japan–India component therefore focuses on adaptation principles rather than technology replication.

Digital-Health Infrastructure in India

India's digital-health backbone centers on the Ayushman Bharat Digital Mission (ABDM), launched in 2021 to build an integrated, federated health ecosystem connecting patients, providers, and facilities nationwide.

ABDM rests on a "trinity of registries", ABHA (patient health IDs), HPR (professionals), and HFR (facilities), enabling consent-based data exchange under the Digital Personal Data Protection Act, 2023.

As of early 2026, ABDM has crossed 78 crore ABHA accounts and 4 crore linked health records, making it one of the largest digital-health rollouts in the world.

An academic assessment found strong progress in beneficiary registration and facility verification but flagged significant inter-state variation in implementation.

A 2025 policy review notes that ABDM has introduced offline and assisted-registration modes for underserved regions, though durable outcomes such as reduced costs and improved health results remain unproven.

Following India's Digital Public Infrastructure model, the government builds core rails while private players innovate on top; over 200 partners, including EkaCare and Bajaj Finserv Health, now build on the ABDM stack.

A 2026 policy explainer frames ABDM's unfinished agenda as consent-UX hardening and ensuring the infrastructure genuinely serves the most vulnerable, directly relevant to a human-centered AI design argument.

Literature Synthesis Matrix

Literature area	Key finding	Implication for proposed research
Indian ageing	India's older population is increasing rapidly, and elderly-care challenges span health, social, and institutional dimensions.	Design for community-level care, affordability, and caregiver involvement.
Rural telehealth / eHealth	Telehealth can improve access, but usability and digital-literacy barriers remain.	Use simple workflows, voice, icons, minimal typing, and assisted onboarding.
Fall detection	Wearable and inertial sensors can detect or predict fall-related events, but real-world validation and data limitations remain.	Compare lightweight models and evaluate false alerts, missing data, and real-world feasibility.
Explainable AI	Explainability can support appropriate trust, but poor explanations may confuse users or create over-trust.	Provide concise, evidence-based explanations and retain human verification.
Japan care technology	Japan promotes technology to improve care quality, reduce staff burden, and support independence.	Adapt workflow and caregiver-support principles rather than copy Japanese technology.
Indian digital health	ABDM promotes interoperable digital health with privacy and consent mechanisms.	Explore standards-based integration subject to authorization and technical feasibility.

Research Gap Analysis

The literature reviewed across six areas shows that the individual components of a human-centered elderly-care AI system already exist, but nowhere are they integrated for India's rural elderly. The core gap is that infrastructure, detection AI, and explainability are studied as separate tracks; no study integrates them into a single, human-centered, co-designed system for this population, which is the specific contribution this proposal aims to make.

Research on Indian ageing quantifies the demographic urgency (NPHCE projects roughly 12% elderly population) and digital exclusion (fewer than 10% of seniors transact online), but stops short of translating this into concrete design requirements for AI interfaces. Studies of rural telehealth and eHealth show that platforms such as

eSanjeevani have scaled to more than 163 million consultations, yet usage skews toward the 25–45 age group; the platform was not designed with elderly users as a primary target. Fall-detection research is technically mature, with sensor-based models achieving 97–99% accuracy, including validated Indian field studies from Pune, but it remains an isolated technical solution that has not been integrated into a broader remote-monitoring system. Explainable AI is shown to increase elderly users' trust and reduce cognitive load, but only in Western and East Asian studies; no research has yet tested explainability for low-literacy, regional-language Indian users. Japan's care-technology sector offers a cautionary lesson: despite heavy R&D investment, its gerontechnology has often followed a paternalistic design approach that excludes elderly input. Finally, India's digital-health infrastructure (ABDM) is robust at the policy level, with 78 crore ABHA IDs issued, but remains untested for actual usability by elderly, low-literacy users.

Problem Statement

Current remote-care applications are often designed for digitally literate users and may not adequately address the needs of elderly people in rural or semi-urban India. Many systems also generate raw sensor data without giving caregivers actionable, prioritized information.

Central problem: how can a low-cost, human-centered, and privacy-preserving AI system use ordinary smartphones and wearable sensors to detect health risks among elderly people and provide understandable, timely, and culturally appropriate support to caregivers and healthcare workers?

Aim

To design, develop, and evaluate a human-centered AI platform for affordable remote monitoring of elderly people in rural and semi-urban India, inspired by Japanese approaches to elderly-care technology, human-machine interaction, and privacy-conscious health-data management.

Objectives

This study aims to design, build, and pilot a human-centered AI system for elderly care and remote health monitoring tailored to Indian conditions. The specific objectives are:

Understand user needs: study what elderly users, family caregivers, community health workers, and clinicians actually need from the system.

Build fall and movement detection: develop AI that detects falls and unusual movement patterns.

Monitor vital signs: track abnormal heart rate, oxygen levels, blood pressure, or temperature using compatible devices.

Spot behavioral changes: detect prolonged inactivity or shifts in daily routine.

Track medication adherence: support monitoring of possible missed medications.

Build a caregiver dashboard: show risk scores, prioritized alerts, patient history, and suggested next steps.

Design a simple interface: create a multilingual UI using voice, icons, large text, and local-language prompts.

Learn from Japan: study Japanese elderly-care technology practices adaptable to Indian conditions.

Run a pilot program: test the system with hospitals, NGOs, or elderly-care centers across two or three districts.

Evaluate the system: assess accuracy, usability, affordability, privacy, reliability, and integration potential with India's digital-health infrastructure (such as ABDM).

Research Questions

The central research question is: how effectively can a human-centered AI system detect and communicate health risks among elderly users in rural and semi-urban India? The following sub-questions address specific components of this question.

On the technical side, the study asks which combinations of smartphone and wearable-sensor data reliably support fall and abnormal-vital detection, and whether machine-learning models trained on this data can generate risk scores that are useful without producing excessive false alarms. It also asks what privacy-preserving architecture is appropriate for settings with unreliable connectivity, a constraint that shapes most other design decisions.

On the human side, the study examines how elderly users actually engage with voice-based, icon-based, and vernacular interfaces, rather than how designers assume they will. It also asks how caregivers perceive the dashboard: whether it is useful, trustworthy, and worth the additional step in their day, or whether it adds unnecessary workload. Finally, given Japan's decades of experience with elderly-care technology, the study asks which human-machine interaction principles from that experience might transfer to Indian conditions, and which would not.

Proposed System

The system is organized into four layers, moving from raw data collection to the interfaces people actually use: the sensor layer, the AI analytics layer, the caregiver dashboard, and the elderly user interface.

Sensor Layer

Smartphone accelerometer and gyroscope.

Smartwatch or fitness-band movement sensors.

Heart-rate and oxygen-saturation sensors.

Blood-pressure and temperature devices, where available.

GPS or location data only with explicit consent.

Medication reminders and caregiver-confirmed adherence records.

Manual symptom reporting through voice, icons, or simple questions.

The system will support a minimum-device mode, allowing users without smartwatches to rely primarily on a smartphone and periodic manual measurements.

AI Analytics Layer

- Machine-learning classification for fall detection.
- Time-series anomaly detection for unusual activity or vital patterns.
- Behavioural-baseline modelling for changes in routine.
- Medication-adherence prediction based on reminder responses and caregiver confirmation.

The system will not diagnose disease. It will instead categorize cases as low risk, moderate risk, or high risk. High-risk alerts will require human verification.

Caregiver Dashboard

- Patient list with risk categories.
- Real-time and historical alerts.
- Vital-sign trends.
- Fall and inactivity events.
- Medication reminder status.
- Last successful device synchronization.
- Escalation workflow for family members, community health workers, and clinicians.
- Notes and follow-up records.

- Offline data synchronization.

Alerts will be ranked by urgency and grouped to reduce unnecessary notifications.

Elderly User Interface

- Marathi, Hindi, and English options, with scope for additional local languages.
- Large buttons and high-contrast visual elements.
- Voice instructions and speech-based responses.
- Minimal typing.
- Familiar icons for medicine, emergency, doctor, and family.
- One-button emergency calling.
- Confirmation prompts after a fall or abnormal reading.
- Offline operation with delayed synchronization.
- Family or community-worker assistance during initial setup.

Methodology

The study will use a mixed-methods, design-science research approach across five phases: requirement analysis; system design and prototype development; AI-model development and validation; field-pilot implementation; and evaluation and refinement.

Phase One: Requirement Analysis

Semi-structured interviews and focus-group discussions will be conducted with elderly users, family caregivers, community health workers, nurses and doctors, NGO representatives, and technology/public-health experts.

The study will investigate healthcare needs, common emergency situations, technology barriers, preferred languages, concerns about surveillance, and acceptable alert mechanisms.

Phase Two: System Design

The prototype will be designed using participatory methods. Elderly participants and caregivers will review interface layouts, language and voice prompts, alert messages, risk-score explanations, emergency-escalation procedures, and consent/privacy settings.

The design process will follow an iterative cycle: prototype, test, collect feedback, revise, retest.

Phase Three: AI Development

A controlled dataset may be created using public fall-detection datasets, smartphone and wearable sensor data from consenting participants, simulated and supervised fall scenarios, validated vital-sign measurements, and medication-reminder interaction logs.

The research will compare decision trees, random forests, support-vector machines, gradient-boosting models, and lightweight neural networks. Model selection will consider accuracy, battery use, computational cost, interpretability, and performance under missing or noisy data.

Phase Four: Field Pilot

One district near a major urban healthcare facility.

One rural district with limited specialist access.

One semi-urban district with community-health infrastructure.

Potential implementation partners include district hospitals, primary health centres, NGOs, senior-citizen associations, home-care providers, and community health workers. The pilot may run for approximately 8–12 weeks, depending on ethical approval, recruitment, and technical readiness.

Phase Five: Evaluation

Evaluation area	Key measures
Technical performance	Fall-detection accuracy; sensitivity and specificity; false-alert rate; abnormal-vital detection; medication-adherence classification; alert delivery time; battery/data usage; system uptime; performance under network interruption.
Human and clinical usability	System Usability Scale score; elderly task completion; caregiver satisfaction; alert understandability; trust; perceived caregiver workload; acceptance of voice/vernacular features; correctly acted-upon alerts.
Economic and deployment feasibility	Approximate cost per user; device/connectivity requirements; training time; maintenance; scalability; compatibility with ABDM and e-Sanjeevani workflows.

Data Privacy and Ethics

Working with elderly patients' health data means that privacy cannot be treated as an afterthought; it must shape the design from the outset. This is especially important in a caregiving context, where the person sharing the data is not always the person who fully understands what happens to it afterward.

The following principles will guide data handling:

Informed consent: consent will be explained in the participant's own language and obtained from both the elderly participant and their caregiver where appropriate. Consent given without genuine comprehension is not considered valid consent.

Data minimization: information such as location will be collected only with explicit user agreement, and no additional data will be collected beyond what is needed.

Data protection: data will be encrypted in storage and in transit, with role-based access so that a family member sees different information than a clinician. Research datasets will be pseudonymized to limit re-identification.

Transparency: users will be able to see why a given risk score was assigned rather than receiving a number alone, and a clear policy will govern how long data is retained and when it is deleted.

Human oversight: a human will review the system's outputs, particularly for high-stakes cases, with a backup plan for participants who cannot respond to an alert themselves.

This approach follows a basic principle in medical AI ethics: the system should function as decision support, not decision authority. It can flag, sort, and prioritize risk, but it should not have the final word on a person's health. A high-risk alert will always trigger a human response, such as a phone call, home visit, or clinical consultation, before any action is taken. The AI narrows down where attention is needed; a person decides what that attention involves.

Integration with Indian Health Systems

The platform is intended to complement, rather than duplicate, existing public-health infrastructure. ABDM provides an interoperable digital-health ecosystem and supports consent-managed health information, while e-Sanjeevani provides a pathway for remote consultation. Actual integration will depend on authorization, standards, technical access, and applicable governance requirements. Planned integration points include:

Linking a participant profile to an ABHA account where appropriate and authorized.

Referring high-risk cases to e-Sanjeevani consultations.

Supporting the National Programme for Health Care of the Elderly.

Providing community-health workers with follow-up lists.

Generating anonymized district-level reports for planning.

Supporting relevant Ayushman Bharat programmes for eligible older adults.

Expected Outcomes

The outcomes of this project fall into three categories: what will be built, what will be learned, and what will be written.

What will be built:

A working prototype of the full elderly remote-monitoring system.

Lightweight AI models for detecting falls, tracking vital signs, and flagging possible medication non-adherence, designed to run on modest hardware rather than hospital-grade infrastructure.

A caregiver dashboard that shows risk scores alongside a plain-language explanation of why each score was assigned.

A multilingual, voice-enabled interface for elderly users that does not require typing or reading small text.

A privacy-preserving architecture designed to continue functioning where internet connectivity is unreliable.

What will be learned:

Empirical usability and acceptance findings on how Indian elderly users and their caregivers respond to the system in practice.

A Japan–India comparative framework identifying which aspects of Japan's long-standing elderly-care technology experience are transferable to Indian conditions, and which are not.

Practical recommendations for integrating with ABDM, e-Sanjeevani, and district-level healthcare systems, since interoperability with existing infrastructure will affect how far the system can scale.

A feasibility assessment for expanding the system beyond the pilot districts.

What will be written:

The full academic output: an MCA dissertation, a research paper, and technical implementation documentation, so that the work can be reviewed and built on by others.

Scope and Limitations

Scope

This project addresses early risk detection and care coordination, not medical diagnosis. It is intended to flag when a situation may need attention, not to identify a specific medical condition. To remain manageable, the initial scope is narrow: falls, abnormal vital signs, prolonged inactivity, and medication reminders. A broader range of conditions may be considered in future work, once this initial set has been validated.

Limitations

Several practical constraints are anticipated and are noted here rather than left to emerge during the project:

- Sensor inaccuracies and variation between devices.
- Inconsistent internet connectivity in the target areas.
- Low smartphone ownership among some elderly participants.
- False positives, where the system flags something as unusual that is in fact harmless.
- Difficulty collecting sufficient real-world fall data, since falls do not occur on a schedule.

- Language and dialect differences across regions.
- Delays in caregiver response to alerts.
- Limited generalizability, given that the pilot covers only two or three districts.
- Dependence on participants consenting to, and continuing to use, the devices over time.

Indicative Budget Heads

A detailed budget will depend on the selected districts, participants, devices, and implementation partners. Major cost heads may include:

Smartphones, wearable devices, and compatible vital-sign measurement devices.

Cloud/server infrastructure and secure data storage.

Software development, testing, and maintenance.

Participant recruitment, travel, and field-study expenses.

Training and onboarding of elderly users, caregivers, and community workers.

Ethics review, documentation, and research administration.

Data analysis and evaluation.

Dissemination, publication, and dissertation preparation.

Research Significance

This research aims to bridge the gap between technically capable AI systems and the practical realities of elderly care in rural and semi-urban India. Its contribution is expected to be both technical and human-centered: developing lightweight risk-detection methods while ensuring that alerts, interfaces, consent mechanisms, and workflows remain understandable and useful to the people who depend on them.

The Japan–India comparative dimension is intended to provide a structured understanding of which human-centered elderly-care technologies and deployment principles are transferable, which require adaptation, and which may not be appropriate in Indian settings.

Conclusion

This research proposes a privacy-preserving, multilingual platform to support remote elderly care in rural and semi-urban India. Early fieldwork points to real demand for something like this. In conversations with three or four elderly residents of the researcher's village along with their sons and grandsons the same issues kept surfacing: joint and muscle pain, unsteadiness on their feet, mood changes, long stretches of inactivity, and habits like tobacco use that nobody in the family seemed able to address.

Most of the elderly participants already owned smartphones, and nearly all of them said they worried about their own health in ways nobody was tracking. None had used an AI-based care system before, but most were open to trying one especially if it gave their children or grandchildren some visibility too. Every single person interviewed said the same thing on language: it had to work in their own dialect, not just Hindi or English.

What became clearer through these conversations was less about technology gaps and more about time. Families want to check in constantly and can't. A son working in another city can't watch for a fall or notice his father skipped his blood pressure check for the third day running. So the system this research is building toward isn't meant to replace a family member or a doctor it's meant to notice things a distracted, distant family can't, and flag them. Phone sensors and wearables, paired with AI that can actually explain its reasoning rather than just alarm, seem like the right starting point provided a human still makes the final call.

Japan's approach to elderly care offers a useful contrast here: less about surveillance, more about keeping people active, social, and independent. That framing matters, but it can't just be imported. India's infrastructure, affordability constraints, language diversity, and reliance on multi-generational households mean the system has to be built around what's actually here, not adapted from what worked somewhere else.

Right now this work is still in the design and review phase. What's left to figure out: which sensors are both affordable and appropriate, which AI methods actually hold up, how to handle privacy properly, and what a realistic deployment even looks like in a village setting. The next steps are AI validation, participatory design sessions, usability testing, and eventually a small field pilot. If it works, the goal isn't a monitoring system that watches elderly people it's one that helps families and health workers stay involved without needing to be everywhere at once.

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Credit and Attribution

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This document is a structured research proposal based on the supplied research concept, methodology, timeline, and Japan–India adaptation framework, with an expanded literature review and corrected tables. The literature review incorporates evidence from peer-reviewed studies and official institutional sources identified during preparation of this revised document.